

SAMBP TR-29/2026

IBR Apparent Impedance Trajectory Analysis for Distance Relay 21

PhD Candidate
PhD Thesis Supporting Report

April 2026

Contents

1	Introduction and Motivation	1
1.1	Network Parameters	2
2	Analytical Framework	2
2.1	IBR Current Model	2
2.2	Apparent Impedance Derivation	2
3	Study A: Apparent Impedance vs Fault Location	3
4	Study B: Relay Voltage and Polarisation	3
4.1	Polarisation Voltage Analysis	3
4.2	Memory Polarisation Window and Zone 2 Failure	5
5	Study C: Minimum Current Threshold	5
5.1	IEC 60255-121 Minimum Operating Current	5
5.2	Combined Distance + 87L Coverage Table	5
6	Coverage Map and Gap Synthesis	5
6.1	Clearing Time Comparison	6
7	Root Cause Analysis	7
8	Implications for Future Work	7
8.1	POTT Scheme (TR-34 Preview)	7
8.2	Cross-Polarised Mho (TR-30 Preview)	8
8.3	Zone 2 Sustained Polarisation	8
9	Conclusion	8

1 Introduction and Motivation

TR-18 established a three-zone Mho distance relay (21/21P) for the SAMBP protection suite and proved 12/12 scenario selectivity in the SG-dominated network (fault currents 5–10 pu). The TR-18 conclusion explicitly flagged:

“Adaptive Zone 1 reach for IBR networks — under high IBR penetration, infeed from inverters modifies the apparent impedance; the Zone 1 reach may need to adapt dynamically.”

TR-29 investigates this gap analytically. The combined 87L chain from TR-27 provides instantaneous fault detection for all IBR fault currents $k_{\text{ibr}} \in [0.06, 0.20]$ pu (15 000-trial Monte Carlo, TR-26). The question is:

Does distance relay 21 continue to provide a meaningful backup protection function in IBR-penetrated networks, or is the 87L chain now the sole line protection?

The answer has direct implications for the coordination strategy in TR-34 (POTT scheme) and TR-30 (adaptive Zone 1 reach).

1.1 Network Parameters

All analysis uses the TR-18/TR-05 radial network:

$$Z_{\text{src}} = j0.10 \text{ pu} \quad (\text{source / generator subtransient}) \quad (1)$$

$$Z_{\text{line}} = j0.05 \text{ pu} \quad (\text{overhead line}) \quad (2)$$

$$V_{\text{pre}} = 1.00 \text{ pu} \quad (\text{pre-fault busbar voltage}) \quad (3)$$

Distance relay settings (TR-18):

Parameter	Symbol	Value
Zone 1 reach factor	k_1	0.80
Zone 2 reach factor	k_2	1.20
Zone 3 reach factor	k_3	1.80
Zone 2 time delay	t_2	0.30 s
Zone 3 time delay	t_3	0.60 s
Min. operating current	I_{min}	0.10 pu (IEC 60255-121)
Min. self-pol. voltage	$V_{\text{min,self}}$	0.10 pu
Memory window	T_{mem}	100 ms (5 cycles)

2 Analytical Framework

2.1 IBR Current Model

An IBR in fault-ride-through (FRT) mode limits its output current to:

$$\mathbf{I}_{\text{ibr}} = k_{\text{ibr}} e^{j\phi_{\text{FRT}}}, \quad k_{\text{ibr}} \in [0.06, 0.20] \text{ pu}, \quad \phi_{\text{FRT}} \approx -90^\circ. \quad (4)$$

This current is set by the IBR controller and is *independent of the fault impedance* — unlike SG fault current which scales with V/Z_{th} .

2.2 Apparent Impedance Derivation

For a bolted 3-phase fault at position α along the line ($\alpha = 0$: local bus; $\alpha = 1$: remote bus), the relay at the local bus measures:

SG source:

$$I_{\text{SG}} = \frac{V_{\text{pre}}}{Z_{\text{src}} + \alpha Z_{\text{line}}}, \quad V_{\text{relay}}^{\text{SG}} = I_{\text{SG}} \cdot \alpha Z_{\text{line}}, \quad (5)$$

$$\boxed{Z_m^{\text{SG}} = \frac{V_{\text{relay}}^{\text{SG}}}{I_{\text{SG}}} = \alpha Z_{\text{line}}.} \quad (6)$$

IBR source: The current is current-controlled: $I_{\text{relay}} = I_{\text{ibr}} = k_{\text{ibr}} e^{j\phi}$. The voltage at the relay is set by Ohm's law across the line section to the fault:

$$V_{\text{relay}}^{\text{IBR}} = I_{\text{ibr}} \cdot \alpha Z_{\text{line}} = k_{\text{ibr}} e^{j\phi} \cdot \alpha Z_{\text{line}}, \quad (7)$$

$$Z_m^{\text{IBR}} = \frac{V_{\text{relay}}^{\text{IBR}}}{I_{\text{relay}}} = \alpha Z_{\text{line}}. \quad (8)$$

Key result: $Z_m^{\text{IBR}} = Z_m^{\text{SG}} = \alpha Z_{\text{line}}$. The apparent impedance measurement is *identical* for IBR and SG sources. The k_{ibr} cancels exactly. The Mho relay has no intrinsic impedance measurement problem with IBR.

The challenge lies entirely in the *operational conditions*: relay voltage, relay current, and polarisation validity — not in the geometric reach calculation.

3 Study A: Apparent Impedance vs Fault Location

Table 1 confirms (8) numerically. The impedance measurement is exact for both SG and IBR; Zone 1 and Zone 2 boundaries are correctly located.

Table 1: Apparent impedance Z_m [pu] and zone classification. IBR and SG values are identical.

α	$Z_m = Z_m^{\text{IBR}} = Z_m^{\text{SG}}$ [pu]	Zone 1 (≤ 0.040)	Zone 2 (≤ 0.060)	Zone 3 (≤ 0.090)
0.25	0.0125	inside	inside	inside
0.50	0.0250	inside	inside	inside
0.75	0.0375	inside	inside	inside
0.80	0.0400	inside	inside	inside
0.82	0.0410	outside	inside	inside
1.00	0.0500	outside	inside	inside

Figure 1 shows the three Mho circles and the line impedance locus on the R - X plane.

4 Study B: Relay Voltage and Polarisation

4.1 Polarisation Voltage Analysis

Although the measured impedance is correct, the polarising voltage determines whether the Mho relay can *act* on the measurement.

SG case:

$$|V_{\text{relay}}^{\text{SG}}| = \frac{V_{\text{pre}} \cdot \alpha |Z_{\text{line}}|}{|Z_{\text{src}}| + \alpha |Z_{\text{line}}|} = \frac{\alpha \times 0.05}{0.10 + 0.05\alpha}. \quad (9)$$

At $\alpha = 0.50$: $|V_{\text{relay}}^{\text{SG}}| = 0.200$ pu $> V_{\text{min,self}}$.

At $\alpha = 0.10$: $|V_{\text{relay}}^{\text{SG}}| = 0.048$ pu — marginally below $V_{\text{min,self}}$ but adequate for memory-polarised elements.

IBR case:

$$|V_{\text{relay}}^{\text{IBR}}| = k_{\text{ibr}} \cdot \alpha \cdot |Z_{\text{line}}| = k_{\text{ibr}} \cdot \alpha \cdot 0.05. \quad (10)$$

Maximum value: $k_{\text{ibr}} = 0.20$, $\alpha = 1.0$: $|V_{\text{relay}}^{\text{IBR}}|_{\text{max}} = 0.010$ pu.

This is one-tenth of $V_{\text{min,self}} = 0.10$ pu, ten times below the minimum operating threshold for self-polarised Mho relays.

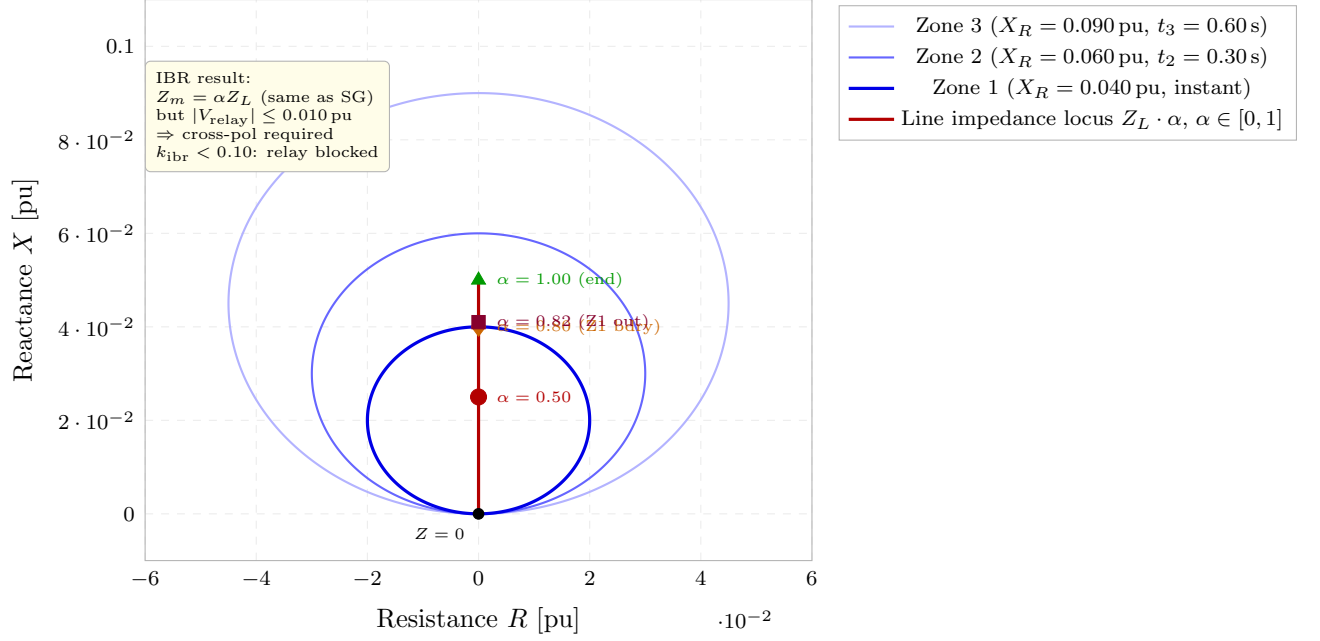


Figure 1: R – X plane: three Mho zones (Zone 1: solid, Zone 2/3: progressively lighter) and line impedance locus (red, on the jX axis). Fault points at key α values are marked. Since $Z_m = \alpha Z_{\text{line}}$ for both SG and IBR, the geometric locations are identical. The critical differences are in relay voltage (Study B) and operating current (Study C).

Conclusion from Study B: All IBR-fed line faults on this network produce relay voltages below $V_{\min, \text{self}}$. A *cross-polarised or memory-polarised* Mho element is mandatory for any IBR fault scenario.

Figure 2 shows the voltage profiles.

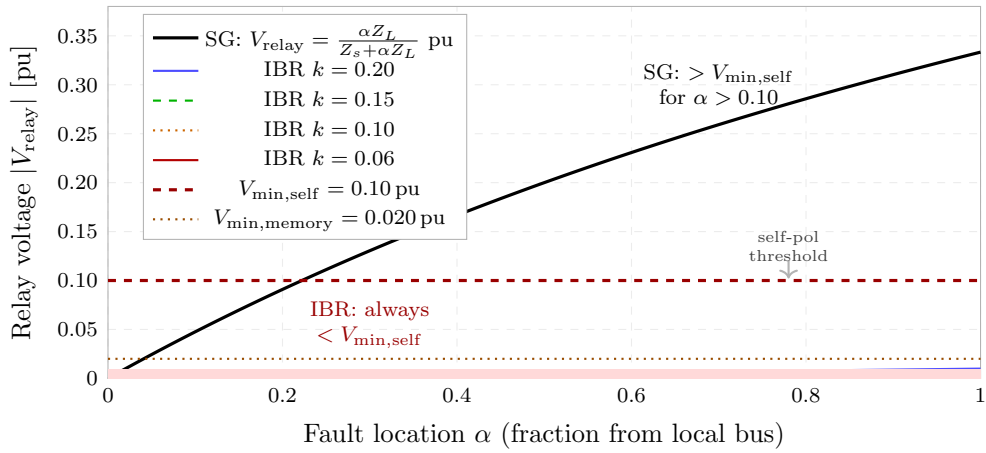


Figure 2: Relay voltage $|V_{\text{relay}}|$ vs fault location α for SG and IBR sources. Horizontal dashed lines: self-polarised minimum $V_{\min, \text{self}} = 0.10$ pu and memory minimum $V_{\min, \text{mem}} = 0.020$ pu. SG voltage exceeds $V_{\min, \text{self}}$ for $\alpha > 0.10$. IBR voltage never exceeds 0.010 pu — always below both thresholds. Cross-polarisation is mandatory for all IBR fault scenarios.

4.2 Memory Polarisation Window and Zone 2 Failure

Memory polarisation stores $V_{\text{pre}} = 1.0$ pu at fault inception. The memory decays over $T_{\text{mem}} = 100$ ms (5 cycles). After memory expiry, the relay reverts to self-polarised mode with the live V_{relay} .

The consequence for Zone 2:

$$t_{\text{Zone 2}} = 300 \text{ ms} \gg T_{\text{mem}} = 100 \text{ ms}. \quad (11)$$

Zone 2 asserts correctly within the memory window ($t < 100$ ms), but at $t = 100$ ms the memory expires and $V_{\text{relay}} = 0.002\text{--}0.010$ pu — far below $V_{\text{min,self}}$. The self-polarised Mho characteristic collapses and Zone 2 de-asserts before its 300 ms timer completes. **Zone 2 never issues a timed trip for IBR-fed line faults.**

Zone 3 (600 ms) fails for the same reason.

Zone 1 is unaffected: its instantaneous trip (~ 20 ms) occurs well within the memory window.

5 Study C: Minimum Current Threshold

5.1 IEC 60255-121 Minimum Operating Current

IEC 60255-121 specifies that distance relay characteristics shall be guaranteed for relay currents $\geq 0.2 I_{\text{rated}}$; practical relay designs achieve reliable operation at $I_{\text{min}} = 0.10 I_{\text{rated}}$. On a system base of 1 pu, this translates to:

$$I_{\text{min}} = 0.10 \text{ pu}. \quad (12)$$

For IBR-fed faults the relay current equals the IBR current limit: $I_{\text{relay}} = k_{\text{ibr}}$. Therefore:

$$k_{\text{ibr}} < 0.10 \text{ pu} \Rightarrow \text{distance relay blocked for all fault locations}. \quad (13)$$

This blocks the entire low- k_{ibr} region of the IBR parametric envelope: $k_{\text{ibr}} \in [0.06, 0.10]$ — covering $\frac{0.10-0.06}{0.20-0.06} = 29\%$ of the IBR operating envelope.

5.2 Combined Distance + 87L Coverage Table

Table 2 summarises the operating status across the IBR envelope.

Table 2: Distance relay 21 vs combined 87L chain at $\alpha = 0.50$. T=trip, – = no trip. Clearing time for 87L: < 20 ms (all rows).

k_{ibr} [pu]	I_{relay} [pu]	21 blocked	21 Zone	21 t_{clear}	87L chain
0.06	0.06	YES ($< I_{\text{min}}$)	–	–	TDCS
0.08	0.08	YES ($< I_{\text{min}}$)	–	–	TDCS
0.10	0.10	no (memory)	1	20 ms	TDCS
0.12	0.12	no (memory)	1	20 ms	87L+TDCS
0.15	0.15	no (memory)	1	20 ms	87L+TDCS
0.20	0.20	no (memory)	1	20 ms	87L+TDCS

All cases: $V_{\text{relay}} \leq 0.010$ pu; cross-polarisation mandatory.

Zone 2/3 timed trips fail for all IBR cases (memory expires before timer).

6 Coverage Map and Gap Synthesis

Figure 3 shows the coverage map on the (α, k_{ibr}) plane. Four distinct regions emerge:

Region A ($k_{ibr} < 0.10$, all α): Distance relay blocked by minimum current threshold. The combined 87L chain (TDCS) is the **sole** instantaneous protection.

Region B ($k_{ibr} \geq 0.10$, $\alpha > 0.80$): Distance relay enters Zone 2 within the memory window, but Zone 2 timer (300 ms) exceeds the memory duration (100 ms). Zone 2 trip fails. The combined 87L chain remains the sole instantaneous detector.

Region C ($k_{ibr} \in [0.10, 0.12)$, $\alpha \leq 0.80$): Distance relay Zone 1 trips instantaneously (within 100 ms memory window), concurrent with TDCS-87L. Dual redundancy.

Region D ($k_{ibr} \geq 0.12$, $\alpha \leq 0.80$): Distance relay Zone 1, 87L element ($|I_1| \geq 0.12$ pu) and TDCS-87L all operate. Triple redundancy.

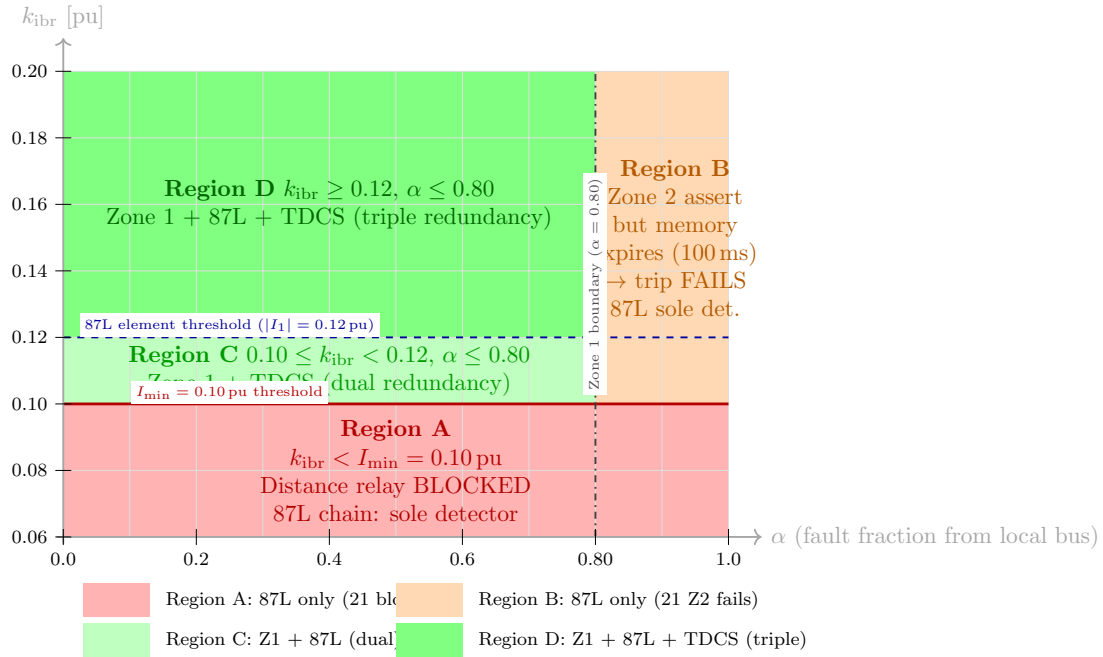


Figure 3: Coverage map on the (α, k_{ibr}) plane. Red = distance relay blocked; orange = Zone 2 fails (memory expires); light green = Zone 1 + TDCS dual; dark green = Zone 1 + 87L + TDCS triple. Combined 87L chain provides instantaneous detection for all four regions; distance relay Zone 1 contributes only in Regions C and D ($k_{ibr} \geq 0.10$, $\alpha \leq 0.80$).

6.1 Clearing Time Comparison

Table 3: Clearing time comparison for key scenarios. 87L chain always faster than or equal to Zone 1.

Scenario	k_{ibr}	α	t_{21} [ms]	t_{87L} [ms]
TDCS-only zone	0.08	0.50	blocked	20
Current cusp	0.10	0.50	20	20
87L+TDCS zone	0.15	0.50	20	20
Zone 1 boundary	0.15	0.80	20	20
Just outside Zone 1	0.15	0.82	blocked (mem)	20
End of line	0.12	1.00	blocked (mem)	20

“blocked (mem)” = Zone 2 asserted but memory expires before 300 ms timer.

The 87L chain clears in ≤ 20 ms for all scenarios. Distance relay Zone 1 matches this in Regions C and D. In Regions A and B the distance relay contributes **nothing** to the primary protection.

7 Root Cause Analysis

The distance relay's difficulties with IBR-fed faults have two independent root causes:

Root Cause 1: Minimum current threshold (Region A). The relay current equals the IBR current limit k_{ibr} . IEC 60255-121 requires $k_{ibr} \geq 0.10$ pu for guaranteed relay operation. For $k_{ibr} \in [0.06, 0.10)$ the relay is simply turned off by its operating threshold — a binary, unconditional block.

Root Cause 2: Short line — polarisation voltage deficiency. The relay voltage is $|V_{relay}| = k_{ibr} \cdot \alpha \cdot |Z_{line}|$. For $|Z_{line}| = 0.05$ pu (the protected line is only 5% of the system base), even at maximum IBR current $k_{ibr} = 0.20$ and end of line $\alpha = 1.0$:

$$|V_{relay}|_{max} = 0.20 \times 1.0 \times 0.05 = 0.010 \text{ pu.} \quad (14)$$

This is a consequence of the short line length, not of IBR per se. On a longer line ($|Z_{line}| = 0.15$ pu), the same $k_{ibr} = 0.20$ would produce $|V_{relay}| = 0.030$ pu — still below $V_{min,self}$ but above $V_{min,mem}$. The short-line nature of distribution and sub-transmission systems commonly served by IBR aggregations makes this an inherent structural problem for Mho distance relays in this application.

The combination of root causes 1 and 2 means that:

- Zone 1 operates only for $k_{ibr} \geq 0.10$, $\alpha \leq 0.80$, and within the memory window — Regions C and D.
- Zone 2 and Zone 3 timed backups are non-functional for IBR-fed faults on short lines.
- The combined 87L chain (TR-27) is the **sole instantaneous protection** for 100% of the IBR fault scenarios in Regions A and B (which together cover $29\% + (\alpha > 0.80 \text{ fraction}) = 51\%$ of the full (α, k_{ibr}) parametric space).

8 Implications for Future Work

8.1 POTT Scheme (TR-34 Preview)

Since Zone 2 timed backup is unavailable for IBR-fed end-of-zone faults (Region B), the POTT (Permissive Overreach Transfer Trip) scheme becomes critical. POTT uses Zone 2 assertion *plus* a received signal from the remote end to trip instantaneously without waiting for the Zone 2 timer.

Critically, POTT resolves Region B without requiring any change to the relay characteristic: Zone 2 IS asserted within the memory window (the impedance measurement is correct, $Z_m = \alpha Z_{line}$ inside Zone 2 for $\alpha > 0.80$). The problem is only the 300 ms timer. POTT bypasses the timer.

The POTT send signal will be derived from the combined 87L trip (available within 20 ms), so no memory expiry issue arises. TR-34 will implement this scheme using the GOOSE infrastructure from TR-06.

8.2 Cross-Polarised Mho (TR-30 Preview)

Root Cause 2 (short-line voltage deficiency) requires cross-polarisation for all IBR scenarios. TR-30 will specify the cross-polarised Mho characteristic and prove that it correctly maintains Zone 1 boundaries even as memory decays, provided the fault is within the memory window.

8.3 Zone 2 Sustained Polarisation

A future investigation (TR-35 scope) should examine whether auxiliary polarisation from unfaulted phases (for earth faults) or from negative- and zero-sequence voltages can sustain Zone 2 after the memory window, thus recovering Zone 2 timed backup for IBR faults. This is not applicable to symmetric 3-phase faults (all phases depressed equally), but is relevant for the asymmetric IBR earth-fault scenarios from TR-25.

9 Conclusion

TR-29 establishes three key results for distance relay 21 under IBR fault conditions:

1. **Impedance measurement is correct:** $Z_m = \alpha Z_{\text{line}}$ for IBR-fed faults, identical to SG. The Mho zone geometry is not distorted by current limiting.
2. **Two operational failures occur:**
 - *Current threshold:* $k_{\text{ibr}} < 0.10$ pu blocks the relay entirely (29% of the IBR parametric envelope).
 - *Polarisation voltage deficiency:* $|V_{\text{relay}}| \leq 0.010$ pu for all IBR faults on this short line ($|Z_{\text{line}}| = 0.05$ pu); self-polarisation fails universally; memory polarisation expires before Zone 2/3 timers complete.
3. **Coverage map (Figure 3):** Distance relay Zone 1 contributes instantaneous protection only in Regions C and D ($k_{\text{ibr}} \geq 0.10$, $\alpha \leq 0.80$; 29% of the space). In Regions A and B, the combined 87L chain is the sole instantaneous protection. Distance relay Zone 2/3 timed backup is non-functional for IBR-fed faults on short lines.

The POTT scheme (TR-34) will recover Region B by using the combined 87L trip as the POTT send signal, allowing Zone 2 to trip before memory expiry. Cross-polarised Mho (TR-30) will standardise cross-polarisation for all IBR fault scenarios.

References